JY063 EVS stimulator: SAFETY LOOP (planning sheet, Rev 2 target)

Block level, no part values are final. Numbers: [DS] datasheet, [H] hand calc, [N] ngspice (archive 4-rev1-design), [DS5] Digitimer DS5 manual, [ASSUME]. Bench prototype, dummy loads only. Companion text: 03_safety_plan/SAFETY_LOOP.md

PC / FPGA GROUND

STIMULATOR (floating) GROUND

FPGA / PC
Red Pitaya Z7020
DAC_CMD, ENREQ,
HEARTBEAT out;
FAULT_N, RAILOK,
ENABLE status in.
Software limits:
sum = 0, |I| <= 5 mA,
max DC duration

ISO
digital
isolator
SPI -> DAC
status <-

DAC_CMD

DAC + CLAMP
command ceiling
D3/D4 zener clamp
+ buffer rails:
~7 mA (G=1),
~9 mA (G=5) [N]
first current bound

CURRENT PUMP
INA149 Howland
+/-100 V rails
1 mA/V
rail shunts RSP/RSN
(OC sense taps)

SERIES LIMITER
DN2540 pair + RLIM
6.4-8.8 mA [N]
passive, no supply,
both polarities
< 10 mA hard limit

S1 STIM SWITCH
PhotoMOS AQY214S
400 V, Toff 0.2 ms
[DS]
opens with LED off
(no HV-side supply)

STIM

PARTICIPANT
(bench: dummy load
1k-13k, R||C)
STIM electrode ->
<- RET electrode
TVS SMBJ100CA
across each line

RET

S2 RETURN SWITCH
PhotoMOS AQY214S
second, independent
disconnect means
(F07: one welded
still breaks loop)

RETURN SHUNT
RSHR + INA
I_RET = 0.2 V/mA
to mismatch, DC
trip and ADC

GND (floating)

I_CMD (delayed reference)

S3 PARKING
Form B PhotoMOS
100k to GND when
ENABLE low: no
charge dump at
switch-on [N]

LED_DRV (S1, S2, S3)

I_SENSE(+/-)

V_ELEC_DIV

RET_SHUNT

PRINCIPLES
1. Single fault safe: no one failure puts > 10 mA or sustained DC into the participant (client rule: imagine everything shorted).
2. Passive first: limiter and clamp need no supply, no logic (purple).
3. De-energise to safe: the switches need the LED lit to conduct; any lost term opens.
4. Two independent disconnect means in the loop (S1, S2).
5. Trips are analog hardware; firmware only requests, never enables.
6. Latched faults need a human at the box to clear.

+HV / -HV / aux
(from power board J1)

RAIL SUPERVISION
+/-HV UV/OV window
+ symmetry [H]
aux +/-12, +/-9 V
POR 20 ms [DS]
-> RAILOK

OVER-CURRENT TRIP
rail shunts -> 2x INA149
-> TLV1704 window
Itrip ~5.7 mA [N]
catches Q1/Q2 short
to a rail (F03)

COMPLIANCE / OV TRIP
V_elec via 20M/200k
divider, 77-83 V [N]
lifted electrode,
stage saturation
(F05/F06)

DC / CHARGE TRIP
integrator on I_RET
baseline > 50 uA avg
or |Q| > Q_max
[DS5 AvL, ASSUME]
skin burn (F08/F09)

MISMATCH TRIP
|I_RET - I_CMD|
> 0.4 mA, 47 us [N]
detached return
(F05); D-19 open for
shared return

OC

OV

DC

MM

LED_DRV

ISO
digital
isolator
ENREQ ->
HEARTBEAT
->
FAULT_N <-
RAILOK <-

TRIP OR
open-drain wired-OR
TRIP_N = OC | OV |
DC | MM | !RAILOK
any one trips

TRIP_N

FAULT LATCH
SR, set-dominant
(74LVC1G74, D-10)
POR sets at power-up
RESET accepted only if
ENREQ=0 & RAILOK=1
-> FAULT_N (latched)

FAULT_N

ENABLE AND (HARDWIRED)
ENABLE = ENREQ & FAULT_N & RAILOK
& WD_OK & ARM & ENGAGE
74LVC gates, no firmware in the path
loss of any term: de-energise, not latched
(FPGA clears DAC first, then ramps)

ENABLE

SWITCH DRIVER
LEDs of S1, S2, S3 in
series from +12 V
ENABLE high: S1, S2
closed, S3 open
ENABLE low: S1, S2
open, S3 parks

INDICATORS
ARMED (yellow)
ENABLED (green)
FAULT (red, latched)
monitor BNC I, V
(iso amp, PC gnd)

RAILOK

ENREQ

FAULT_N, RAILOK, ENABLE -> FPGA (status, read-only)

WD_OK

RESET

ARM

ENGAGE

LEGEND
red stimulating path
purple passive limit (no supply)
amber detector / trip
blue latch + enable chain
green isolation crossing
thick current path
dashed reference / status

Rev 1 board has: clamp, pump, limiter (DNP), S1, S2, RSHR, read-back. Everything else on this sheet is Rev 2.

HEARTBEAT

WATCHDOG
HEARTBEAT from FPGA
-> missing-pulse
detector -> WD_OK
(F01: FPGA hung or
link lost)

RESET PUSH-BUTTON
on the enclosure, hardware
only; clears the latch; no
software reset path
(DS5: toggle down = reset)

ARM
operator key
switch, latching
(DS5 'Output
On/Off-Reset')

ENGAGE
participant hand
switch, NO,
momentary
(dead-man)

TRIP SUMMARY (TARGETS, TO BE CONFIRMED ON THE BENCH)

OC	|I_rail| > ~5.7 mA, ~22 us	latched
OV	|V_elec| > 77-83 V	latched
DC	avg I_RET > 50 uA (idle) / Q > Q_max	latched
MM	|I_RET - I_CMD| > 0.4 mA, ~47 us	latched
RAIL	+/-HV out of 90-110 V or asymmetric	not latched
WD	no HEARTBEAT edge for > 10 ms [ASSUME]	not latched
ENGAGE / ARM / ENREQ	released	not latched

ISOLATION BARRIER (Rev 2: >= 1 MOPP reference)

Sources: archive 4-rev1-design/DECISIONS.md, FMEA.md, DS5 manual
Bench prototype, dummy loads only; not a safety sign-off
Author: Weymen Koo. Text: 03_safety_plan/SAFETY_LOOP.md
DRAFT: planning sheet, not for fabrication
UBC ELEC 491 JY063

Sheet: /
File: safety-loop.kicad_sch

Title: Safety loop: block schematic (planning)

Size: A3	Date: 2026-11-04	**Rev: 0.1 DRAFT**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1